

Muhammad Adil Usmani

03114086626 | muhammadaadilusmani@gmail.com | linkedin.com/in/muhammad-adil-usmani | github.com/AadilUsmani | v0-muhammadaadilusmani.vercel.app

SUMMARY

Software Engineer specializing in Applicational AI, Retrieval-Augmented Generation (RAG) pipelines, and LLM workflow orchestration. Builds and integrates ML models served through FastAPI across unstructured text and structured JSON data flows, applying prompt and context engineering to optimize LLM outputs across varied retrieval architectures.

TECHNICAL SKILLS

Languages: Python, SQL | **Backend:** FastAPI

AI & LLM: RAG, Prompt Engineering, Context Engineering, LLM Orchestration, LangChain, LangGraph, LangSmith, Azure OpenAI, OpenAI API, Gemini API, GPT-4o-mini, T5 Transformers

Databases: Neo4j, ChromaDB, FAISS, SQLite

Cloud & Tools: Azure, GCP, Render, Vercel, Claude, Claude Code, GitHub Copilot CLI, Digital Ocean, GitHub Actions, Git, Modal, Hugging Face

EXPERIENCE

Data Science Intern — ML1

Jul 2026 – Present

- Build end-to-end automation pipelines and workflows for international clients, including HR process automation and legal data acquisition for applicational AI; developed an internal scheduling tool.

Data Science Trainee — ML1

Jun 2026 – Jul 2026

- Strengthened foundational machine learning knowledge with a focus on implementing and training algorithms from scratch.

RESEARCH

Anarchist LLM — Emergent Pattern Discovery

Python, Modal, GPT-1900

github.com/AadilUsmani/Anarchist-LLM

- Ran inference on GPT-1900 via Modal to evaluate the model's ability to identify and optimize algorithms and patterns without being explicitly told about them, probing emergent unsupervised reasoning behavior.

Implementing Titan Architecture

Python, Titan Architecture

github.com/AadilUsmani/implementing_titan_architecture

- Built backend infrastructure using the Titan architecture to process large-scale financial datasets (incl. Engro 2025 annual report) with PDF extraction and preprocessing pipelines for corporate documents.

Lexical Graph RAG

Python, Neo4j, LangGraph

github.com/AadilUsmani/Lexical_Graph_RAG

- Engineered a graph-based RAG system with knowledge graph deduplication for unstructured SEC filings; evaluated with Signal-to-Noise and Cumulative Recall metrics to balance latency and hallucination rates.

PROJECTS

AeroSphere – Air Quality Forecasting System

Python, LSTM, Azure, Airflow, NASA TEMPO

aerosphere.earth

- Built a 72-hour PM2.5 forecasting pipeline across 45 cities using LSTM models and NASA TEMPO satellite data via Apache Airflow; integrated GPT-based NLG reporting and deployed the backend on Azure.

Hybrid Graph RAG vs Vector RAG Evaluation System

Python, ChromaDB, Neo4j, Azure OpenAI

github.com/AadilUsmani/V.G.RAG

- Built a hybrid RAG system over Apple/Tesla SEC 10-K filings (ChromaDB + Neo4j) with late-fusion aggregation and a 53-question automated benchmark for accuracy, comprehensiveness, and directness.

Corrective RAG (CRAG) – Adaptive Routing API

Python, LangGraph, FAISS, GPT-4o-mini

github.com/AadilUsmani/Corrective_rag_CRAG

- Architected a LangGraph stateful workflow with confidence-score routing: low-confidence queries (<0.3) trigger live web search; high-confidence (≥0.7) resolve from a local FAISS store.

EDUCATION

University of Central Punjab, Lahore, Pakistan — BS Computer Science

Unique Group of Institutions, Lahore, Pakistan — Intermediate

Lahore Grammar School, Lahore, Pakistan — Schooling

CERTIFICATIONS

Supervised Machine Learning: Regression and Classification | Introduction to Generative AI | Machine Learning Specialization – Stanford University, Coursera (In Progress) | LLMOps Specialization – Duke University, Coursera (In Progress)